

Sam Spiro

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Positions

- 2025– **Assistant Professor**, *Georgia State University*
2022–2025 **NSF Postdoc**, *Rutgers University*
Supervisor: Bhargav Narayanan

Education

- 2017–2022 **Ph.D. in Mathematics**, *UC San Diego (UCSD)*
Adviser: Jacques Verstraëte
2013–2017 **B.S. in Mathematics and Computer Science**, *University of Miami*
2016 (Fall) Budapest Semesters in Mathematics.

Research Interests

I am interested in a wide range of topics within combinatorics including extremal, probabilistic, and enumerative combinatorics. I particularly enjoy problems related to F -free graphs, probabilistic analogs of extremal results, descents and Eulerian numbers, and spectral graph theory.

List of Papers

The following is a partial list of papers; a full list can be found at binatorics.com.

Particularly significant publications are marked by ♠'s.

Projects I proposed at a workshop aimed at students are marked by ◇'s.

1. ♠ **Helly Theorems for Generalized Turán Problems** (with Sean English).
Submitted, 2026.
2. **Random Turán Problems for Graphs with a Vertex Complete to One Part** (with Sean Longbrake).
Submitted, 2026.
3. **Metric Dimensions of March Madness Brackets**.
Submitted, 2026.
4. **Rainbow Trees in Hypercubes** (with Nicholas Crawford, Maya Sankar, and Carl Schildkraut).
Submitted, 2025.
5. **Rational Exponents for General Graphs** (with Sean English).
Submitted, 2025.
6. **On a clique building game of Erdős** (with Alexandru Malekshahian).
Submitted, 2024.
7. **Random Turán Problems for Hypergraph Expansions** (with Jiaxi Nie).
Submitted, 2024.
8. ♠ **Tree Posets** (with Tao Jiang, Sean Longbrake, and Liana Yepremyan).
Canadian Journal of Mathematics 1–32, 2024.

9. **Generalized Quasikernels in Digraphs.**
European Journal of Combinatorics, Volume 133, 2026.
10. ♠ **Sidorenko Hypergraphs and Random Turán Numbers** (with Jiaxi Nie).
Combinatorial Theory, Volume 5 Issue 4, 2025.
11. **Clique Supersaturation** (with Quentin Dubroff, Benjamin Gunby, and Bhargav Narayanan).
SIAM Journal of Discrete Mathematics 39(3), 2025.
12. ♠ **The Random Turán Problem for Theta Graphs** (with Gwen McKinley).
Forum of Mathematics Sigma (Accepted), 2022.
13. ♠ ♦ **Eulerian Polynomials for Digraphs** (with Kyle Celano and Nicholas Sieger).
Combinatorial Theory, Volume 5 Issue 1, 2025. Conference poster FPSAC, 2023.
14. **Random Polynomial Graphs for Random Turán Problems.**
Journal of Graph Theory, Volume 105, Issue 2, 2024.
15. **Incidence-free sets and edge domination in incidence graphs** (with Sam Adriaensen and Sam Mattheus).
Journal of Combinatorial Designs, Volume 32, Issue 2, 2024.
16. **New Eigenvalue Bound for the Fractional Chromatic Number** (with Krystal Guo).
Journal of Graph Theory, Volume 106, Issue 1, 2024.
17. ♦ **Zero Forcing with Random Sets** (with Bryan Curtis, Luyining Gan, Jamie Haddock, and Rachel Lawrence).
Discrete Mathematics, Volume 347, Issue 5, 2024.
18. ♠ **Antichain Codes** (with Benjamin Gunby, Xiaoyu He, and Bhargav Narayanan).
Bulletin of the London Mathematical Society, Volume 55, Issue 6, 2023.
19. ♦ **Semi-restricted Rock, Paper, Scissors** (with Erlang Surya and Ji Zeng).
Electronic Journal of Combinatorics, Volume 30, Issue 4, 2023.
20. **On t -intersecting Hypergraphs with Minimum Positive Codegrees.**
SIAM Journal of Discrete Mathematics, Volume 37-2, 2021.
21. ♦ **Continuously Increasing Subsequences of Random Multiset Permutations** (with Alexander Clifton, Bishal Deb, Yifeng Huang, and Semin Yoo).
European Journal of Combinatorics, Volume 110, 2021. Conference poster FPSAC, 2022.
22. **Maximal Independent Sets in Clique-free Graphs** (with Xiaoyu He and Jiaxi Nie).
European Journal of Combinatorics, Volume 106, 2021.
23. ♦ **Linear Bounds for Cycle-free Saturation Games** (with Sean English, Tomas Masarik, Grace McCourt, Erin Meger, and Michael S. Ross).
The Electronic Journal of Combinatorics 29(3), 2021.
24. ♠ **A Smoother Notion of Spread Hypergraphs.**
Combinatorics Probability and Computing, Volume 32-5, 2021.
25. **The Wiener Index of Signed Graphs.**
Applied Mathematics and Computation, Volume 416, 2021.
26. **Practical Strategies for Card Guessing with Feedback** (with Persi Diaconis and Ron Gra-

ham).

The American Mathematical Monthly, Volume 129 Issue 7, 2021.

27. ♠ **Card Guessing with Partial Feedback** (with Persi Diaconis, Ron Graham, and Xiaoyu He).
Combinatorics, Probability and Computing, 1–20, 2021.
28. **An Averaging Process on Hypergraphs**.
Journal of Applied Probability 1–10, 2020.
29. **Slow Recurrences**.
Journal of Number Theory 218, 370–401, 2020.
30. **Slow Fibonacci Walks** (with Fan Chung and Ron Graham).
Journal of Number Theory 210, 142–170, 2019.

██████████ Fellowships and Grants

2022–2025 NSF Mathematical Sciences Postdoctoral Research Fellowship; DMS-2202730.

2019–2022 NSF Graduate Research Fellowship; DGE-1650112.

2022 NSF Grant for the Graduate Student Combinatorics Conference; DMS-1933360.

██████████ Teaching Experience

Instructor:

Spring 2026 Graduate Graph Theory. GSU

Fall 2025 Introduction to Probability and Statistics. GSU

Fall 2024 Calculus 1 for Life Sciences. Rutgers

Fall 2023 Introduction to Topology. Rutgers

TA:

Spring 2019 Combinatorics. UCSD

Winter 2019 Complex Analysis. UCSD

Fall 2018 Partial Differential Equations. UCSD

Fall 2018 Graduate Linear Algebra. UCSD

Spring 2018 Game Theory. UCSD

Fall 2017 Calculus III. UCSD

██████████ Invited Talks

June 2026 SIAM Conference on Discrete Mathematics, UC San Diego.

April 2026 Combinatorics Seminar, Iowa State University.

April 2026 Triangle Lecture Series, Wake Forest University.

March 2026 Combinatorics Seminar, MIT.

Nov 2025 Atlanta Lecture Series, Emory University.

Sept 2025 Combinatorics Seminar, UNC Wilmington.
 Sept 2025 Discrete Analysis Seminar, Emory University.
 Jan 2025 AMS Special Session on Matrix Analysis and Applications, JMM.
 Oct 2024 Special Session on Advances in Extremal Combinatorics, AMS Fall Western Sectional Meeting.
 Oct 2024 Special Session on Extremal and Structural Graph Theory, AMS Fall Southeastern Sectional Meeting.
 Sept 2024 Combinatorics Seminar, Georgia Tech.
 Sept 2024 Combinatorics Seminar, Emory University.
 Sept 2024 Combinatorics Seminar, University of Southern California.
 April 2024 Special Session on Recent Trends in Graph Theory, AMS Spring Eastern Sectional Meeting.
 Jan 2024 AMS Special Session on Thresholds in Random Structures, JMM.
 June 2023 Minisymposium on Extremal Combinatorics and Beyond, CANADAM.
 April 2022 Special Session on Extremal Graph Theory, AMS Spring Central Sectional Meeting.
 Feb 2023 Workshop on Ramsey Theory, UC San Diego.
 Jan 2023 ILAS - AIM Special Session on The Inverse Eigenvalue Problem for a Graph and Zero Forcing, JMM.
 Jan 2023 AMS Special Session on Research from Graduate Research Workshop in Combinatorics, JMM.
 Oct 2022 Combinatorics Seminar, University of South Carolina.
 Oct 2022 Combinatorics Seminar, University of Delaware.
 Oct 2022 Special Session on Graphs, Hypergraphs, and Set Systems; AMS Fall Western Sectional Meeting.
 May 2022 Combinatorics Seminar, Iowa State University.
 Oct 2021 Combinatorics Seminar, UC Riverside.
 Sept 2021 Graph Theory and Combinatorics Seminar, UIUC.
 July 2021 Minisymposium on Extremal Graph Theory and Combinatorics, SIAM Conference on Discrete Mathematics.
 Oct 2020 Combinatorics Seminar, University of Delaware.

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Refereeing:

2017– Served as referee or mathematical reviewer for over 85 articles with journals including: Random Structures and Algorithms, Combinatorica, American Mathematical Monthly, Advances in Combinatorics, Journal of Combinatorial Theory B, International Mathematics Research Notices, Bulletin of the London Mathematical Society.

Organization:

- 2025– Organizer of GSU combinatorics seminar.
- 2026 Co-organizer of Special Session at AMS Southeastern Sectional.
- 2025 Co-organizer of AMS Special Session at JMM.
- 2024 Co-organizer of Special Session at AMS Western Sectional.
- 2024 Co-organizer of AMS Special Session at JMM.
- 2022–2023 Co-organizer of the Combinatorics-Related Open Problem Seminar. Virtual.
- 2022 Co-organizer of AMS Special Session at JMM.
- 2018–2022 Co-organizer of the Graduate Student Combinatorics Conference 2022. UCSD.
- 2019–2022 Co-organizer of the Graduate Student Seminar. UCSD.
- 2018–2022 Co-organizer of the Graduate Student Combinatorics Seminar. UCSD.
- 2017–2022 Co-organizer of Banach Barski (weekly graduate student social event). UCSD.

Student Mentorship and Outreach:

- 2025– Mentor for GSU's math department High school Internship Program.
- 2026 Faculty mentor for the Graduate Research Workshop in Combinatorics.
- 2025 Speaker at GSU Math and Statistics club.
- 2023 Mentor for independent high school research.
- 2023 Postdoc Mentor for the Graduate Research Workshop in Combinatorics.
- 2022, 2024 Mentor for Directed Reading Program. Rutgers University.
- 2021–2022 Association for Women in Mathematics mentor. UCSD.
- 2020–2022 GradAMP mentor (helped undergraduates apply to graduate school). UCSD.

Expository Notes:

- 2026– Maintained a growing set of 150+ pages of expository notes on my website concerning advanced graph theory topics.
- 2018– Maintained a growing set of 200+ pages of expository notes on my website concerning methods in extremal combinatorics.

--- Experiences

- 2026 Flags in the Mountains Workshop
- 2025 PCMI Research Topic: Probabilistic and Extremal Combinatorics.
- 2025 SLMATH Semester Program on Extremal Combinatorics.
- 2024 AIM Workshop: High-Dimensional Phenomena in Discrete Analysis.
- 2024 BIRS Workshop: Bootstrap Percolation.
- 2024 AIM Square: Crossing Numbers.
- 2022 AMS MRC program: Models and Methods for Sparse (Hyper)Network Science.
- 2021 AMS MRC program: Approaches to Inverse Problems using Combinatorics and Linear Algebra (participant and organizer of special section at JMM 2022).
- 2019 MSRI Graduate Summer School: The Polynomial Method.

2019, 2022 Graduate Research Workshop in Combinatorics.
2016 University of Chicago REU Program.